

Tongue-Tied

How long before all the world communicates like there were just one language?

Man proposed, and God didn't like it: ever since the Tower of Babel, we've been divided by language. Naturally, in the infinitely clichéd "information age," we throw technology at the problem. Computers (notably those of the Google persuasion) have developed machine translation (MT) capabilities, and before you can say a "Hola!", you hear a "Hello." MT has been around much longer than you might think; it's gotten a bit better with time, but note the "bit." Primitive translators would try the word-by-word approach, which didn't go down too well with most people; we now have MT systems that look at grammar and context. Formerly only valued by officious UN officials, the average online person—there's one for five who aren't online—has come to expect Google's "translate this page" link to just work. As you know, it doesn't—at least, not too well.

Who waits any more in these days of instant tech? Online translators' MT engines are expected to do their job on-the-fly, and that they do. (Your mileage will vary.) There's now a WordPress translator plugin you can embed in your Web page to, well, translate. Find out more at <http://tinyurl.com/26pf2a>. "WordPress Translator Plugin of The Future: Translation in 32 languages," it says. Looks to us like an ambitious project—we'll track it. Already extant are mobile applications such as Mobile Translator (EC Edition)—a Symbian application that translates a set number of phrases, and Pocket PROMT 5.0 for Pocket PC and Windows Mobile 5.0. This one costs just \$60, so it can't be too good, we think...

The future as we see it now—with imperfect use of imperfect tools—is of blogs and sites having "language-select" bars embedded; these could become ubiquitous. Take a look at <http://tinyurl.com/23afr3>. It features Applied Language Solutions' translation solution: a translation bar for your Web site.

Translation accuracy depends on the number of words and phrases in the dictionary. Google (yet again!) has a vast quantity of translated information they've collated for analysis, and the dic-

tionary is being added to every day. Again, we'd hedge our bets on Google for coming up with something that'll help us make sense of all those Chinese pages. Think of all the information people on beyond the language divide are missing out on!

Speech-to-speech on-the-fly translation is, naturally, the holy grail. IBM has a project

called MASTOR, for bi-directional English to Mandarin speech-to-speech. "The tight coupling of speech recognition and understanding effectively mitigates the effects of speech recognition errors and non-grammatical inputs...", they claim.

Then there's the International Center for Advanced Communication Technologies (interACT). InterACT director Alex Waibel demonstrated in 2006 "domain-independent, speech-to-speech translation in a lecture, which was simultaneously translated from English to Spanish to German." Quoting Waibel: "In the future, such transducers could be implanted, enabling a speaker to produce any language at will." We think it wouldn't be far-fetched to expect such systems on phones—when it's perfected, that is. (Read: "a long time into the future").

Integrated Wave Technologies has developed a handheld device for US Marines in Iraq, into which a soldier can speak in English—and the device translates it into Iraqi speech or 15 other languages. This one, too, is bi-directional.

It's important to note that translation is a difficult proposition. Like we've mentioned in a different space in this issue, we need better neural networks for better AI, and where is better AI to come from? The NLP (Natural Language Processing) problem is *hard*. It could well be that as applications develop, we might learn to live with their idiosyncrasies. On the Web, people might change their own usage of language—reducing the use of idioms, for example.

We don't see anything like a universal Web language emerging—something like Esperanto. We'll also tell you that, insofar as we're looking into the future, there's not going to be a breakthrough soon. But the language toolbars will be there, for sure, as we see it—and they'll get better and better, slowly, little steps at a time. It's going to be quite a wait, but it'll be worth it. ☒



Illustration: Shrikrishna Patkar



When you put your customer's call on hold,
he can put your **order on hold.**



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